DRG. NO. VCR-001 · SHEET 1 / 4

VCR/AR

[INSTITUTION / DEPARTMENT] · REFRIGERATION & AIR CONDITIONING LAB

SECTION A · THE HARDWARE

A refrigeration rig, taken apart

An original 3D model of a vapour compression refrigeration demonstration rig, exploded into its parts. Numbers on each drawing match the legend.

Explore it in 3D

Scan for the interactive web explorer and AR view.

FIG. 1A · ASSEMBLED RIG

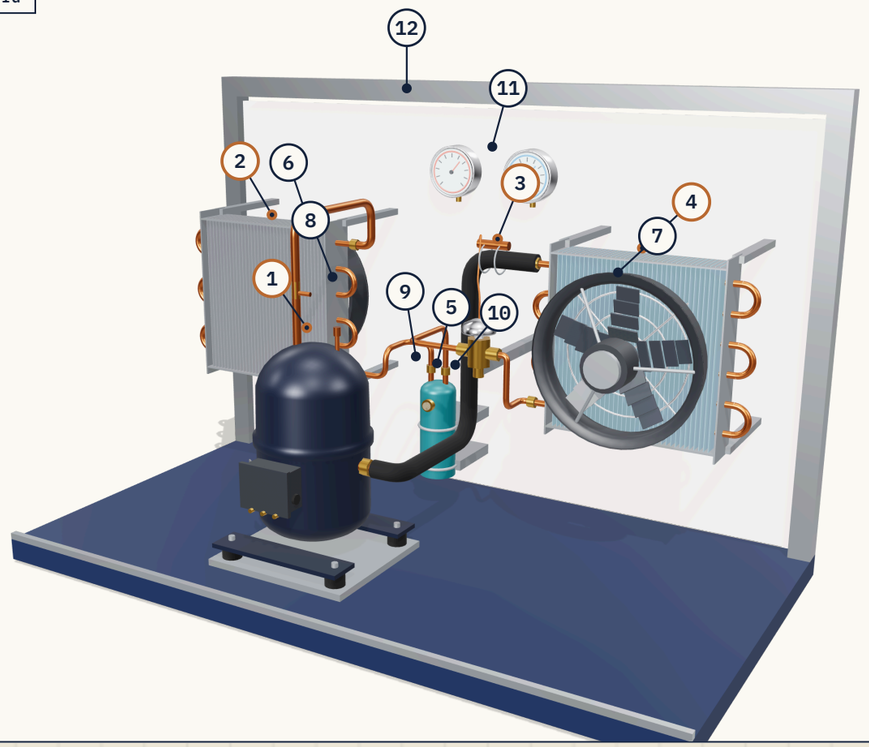
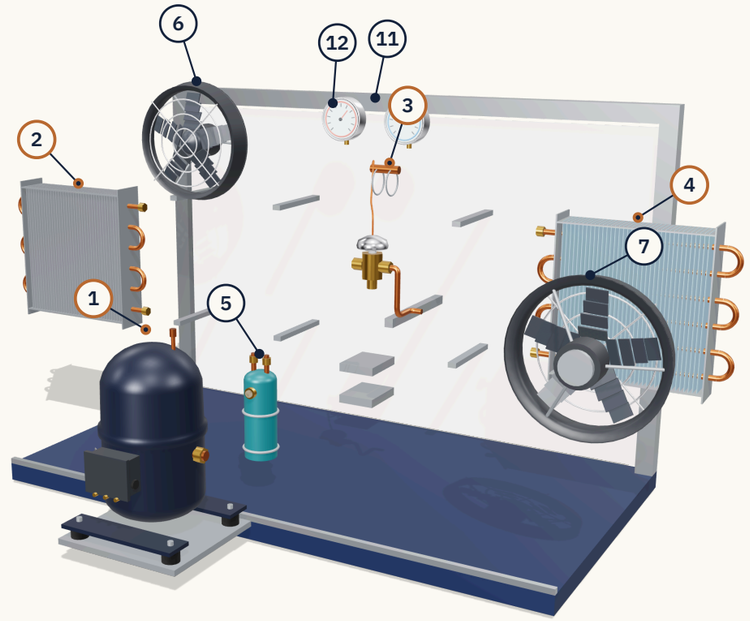


FIG. 1B · EXPLODED VIEW



Component legend

1. **Compressor** · Draws in low-pressure refrigerant vapour from the evaporator and raises its pressure and temperature so it can reject heat in the condenser
2. **Condenser** · Rejects heat from the refrigerant to the surroundings and condenses high-pressure vapour into liquid
3. **Expansion Device** · Throttles the refrigerant, reducing its pressure and generally its temperature, and meters the flow into the evaporator
4. **Evaporator** · Absorbs heat from the space or product being cooled and vaporises the low-pressure refrigerant
5. **Receiver / Filter-Drier** `illustrative` · Stores surplus liquid refrigerant and removes moisture and particles before the expansion device
6. **Condenser Fan** · Forces ambient air through the condenser coil to carry heat away
7. **Evaporator Fan** · Circulates air from the cooled space across the evaporator coil
8. **Discharge Line** · Carries hot, high-pressure vapour from the compressor to the condenser
9. **Liquid Line** · Carries high-pressure liquid refrigerant to the expansion device
10. **Suction Line (insulated)** · Returns low-pressure vapour from the evaporator to the compressor
11. **Pressure Gauges** `illustrative` · Indicate the high-side (condensing) and low-side (evaporating) pressures on a lab trainer
12. **Trainer Frame & Panel** · Supports and positions all components for demonstration

Numbers match the callouts in the web explorer, the Unity app and Sheets 2–4. Core refrigeration cycle parts are numbered 1–4.

Model data verified

MODEL	Original procedural GLB (glTF 2.0), generated by `generate_vcr_glb.py`
VALIDATION	Khronos glTF-Validator: 0 errors, 0 warnings
GEOMETRY	38,700 triangles · 12 named, selectable components
FOOTPRINT	≈ 1.3 × 0.7 m base, ≈ 0.87 m tall (illustrative scale)

Laboratory rig data to be filled

REFRIGERANT	[refrigerant type — to be filled from rig nameplate]
COMPRESSOR	[compressor rating — from nameplate]
CONDENSING P	[measured — bar]
EVAPORATING P	[measured — bar]
COP	[to be calculated from lab readings]

No values are invented: bracketed fields are placeholders for measurements from your own rig.

3D model generated procedurally for this project (original geometry, not CAD, not a specific product). Items 5 and 11 are illustrative accessories. Renders from the project's three.js web viewer.

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THE HARDWARE: VCR RIG, TAKEN APART	1 / 4	NTS

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SECTION B · THE REFRIGERATION CYCLE

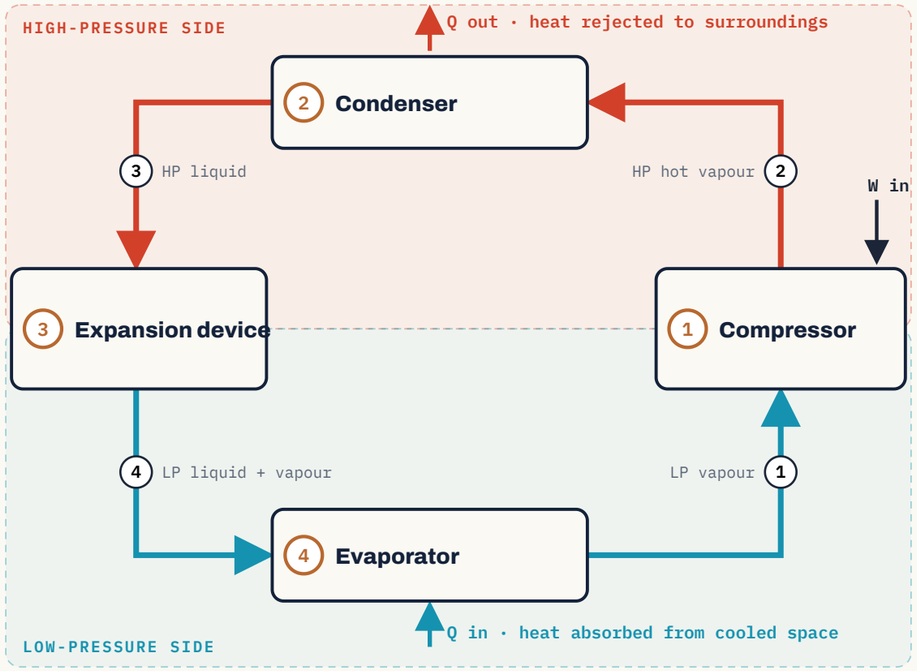
One loop, two pressures

How refrigerant carries heat out of a cold space: it boils at low pressure in the evaporator and condenses at high pressure in the condenser.

Explore it in 3D

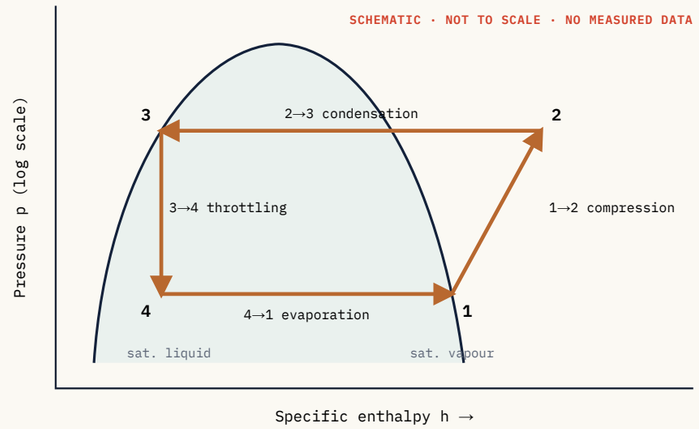
Scan for the interactive web explorer and AR view.

FIG. 2A · CYCLE SCHEMATIC



Typical cycle: Evaporator → Compressor → Condenser → Expansion device → Evaporator. State points 1–4 match Fig. 2B.

FIG. 2B · p-h SKETCH (SCHEMATIC)



High-pressure side

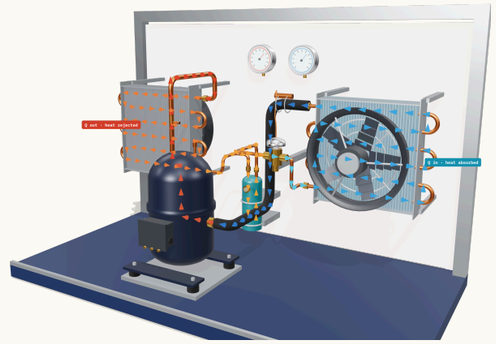
Compressor outlet → condenser → liquid line → expansion device inlet. Refrigerant condenses here and **rejects heat** (Q out) to the surroundings.

Low-pressure side

Expansion device outlet → evaporator → suction line → compressor inlet. Refrigerant boils here and **absorbs heat** (Q in) from the cooled space.

Energy balance (ideal, steady state): Q out = Q in + W in. COP = Q in ÷ W in — calculate it from your own readings.

FIG. 2C · CONCEPTUAL FLOW IN THE 3D MODEL



HP vapour · HP liquid · LP mixture · LP vapour — animation only, not CFD.

1 Evaporation (heat absorbed)

Cold, low-pressure liquid–vapour mixture enters the evaporator. It absorbs heat from the space or product being cooled and boils into low-pressure vapour.

Ideal: Ideal cycle: constant-pressure heat addition until the refrigerant is saturated vapour.

Real: Real systems: a small pressure drop occurs across the coil and the vapour is usually slightly superheated at the outlet so that no liquid reaches the compressor.

2 Compression (work input)

The compressor draws in the low-pressure vapour and squeezes it into high-pressure, high-temperature vapour, using electrical work input.

Ideal: Ideal cycle: reversible adiabatic (isentropic) compression.

Real: Real systems: friction, heat transfer and valve losses make the compression non-isentropic, so more work is needed than in the ideal cycle.

3 Condensation (heat rejected)

Hot, high-pressure vapour gives up heat to the surrounding air through the condenser fins, cools and condenses into high-pressure liquid.

Ideal: Ideal cycle: constant-pressure heat rejection to saturated liquid.

Real: Real systems: the liquid is often slightly subcooled at the outlet, and there is a small frictional pressure drop.

4 Expansion (throttling)

The high-pressure liquid is forced through a restriction. Its pressure falls sharply, part of it flashes into vapour, and its temperature drops before it re-enters the evaporator.

Ideal: Ideal cycle: irreversible throttling at constant enthalpy (no work, no heat transfer).

Real: Real systems: very close to constant enthalpy; a thermostatic expansion valve also adjusts flow to keep evaporator-outlet superheat roughly steady.

Cycle descriptions follow standard refrigeration textbooks (see Sheet 4). Diagrams are schematic and contain no measured laboratory data.

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THE REFRIGERATION CYCLE	2 / 4	NTS

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SECTION C · THE PROJECT

One model, three ways in

Objective, architecture and implementation of the VCR AR Explorer: one procedural model, three ways to learn from it.

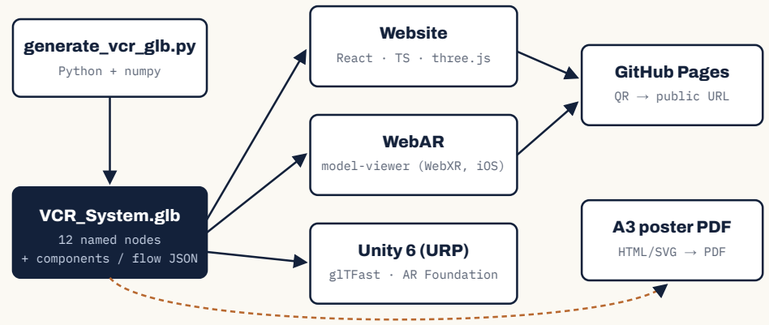
Explore it in 3D
Scan for the interactive web explorer and AR view.

C1 Objective & motivation

Students meet the vapour compression cycle as a diagram, but the real rig in the lab is a tangle of copper. This project connects the two: every part of an original 3D rig can be **tapped to reveal its function**, the refrigerant route is animated stage by stage, and the same model can be placed on a real table in AR — all from one QR code, with nothing to install.

- Identify the four core components and where they sit on a real rig.
- Trace the high- and low-pressure sides and where heat is absorbed or rejected.
- Relate the physical layout to the schematic and p–h sketch on Sheet 2.

C2 Implementation architecture



One shared data source (component text, explode offsets, flow route) drives the website, the Unity app and this poster, so all three always agree.

C3 How it works

3D MODEL
Procedural geometry (lathed shells, swept tubes, fin packs) written straight to glTF 2.0; validated with 0 errors.

WEB EXPLORER
Tap/click picking that ignores drags, single-part highlight, ghost/isolate, explode slider, numbered callouts, cycle player.

AR
Install-free WebAR: WebXR / Scene Viewer on ARCore Android, AR Quick Look on iOS; 3D fallback elsewhere. Camera is requested only when AR starts.

UNITY
Unity 6 + URP: MainMenu and VCRExplorer scenes, ScriptableObject/JSON data, AR Foundation scene for an optional Android build.

FIG. 3A · WEB EXPLORER (DESKTOP)



FIG. 3C · AR PAGE

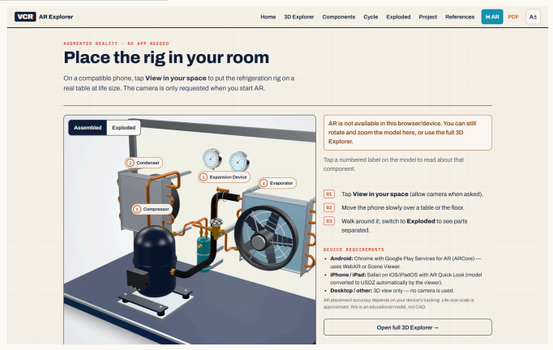


FIG. 3B · PHONE

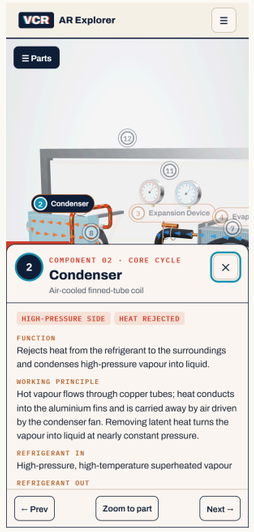
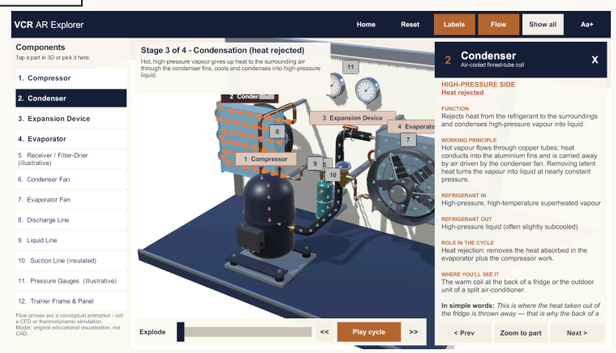


FIG. 3D · UNITY 6 PLAYER · CYCLE MODE



C4 Applications & limitations

Use it for

- Pre-lab briefing before running the real trainer
- Self-study and revision on a phone
- Explaining HVAC systems to non-specialists

Keep in mind

- Educational model — not CAD, not to scale of any product
- Flow animation is conceptual, not CFD or a thermodynamic solver
- No measured data shown; AR depends on the device

Screenshots are captured from the actual web build and Unity player of this project.

TITLE	SHEET	SCALE
THE PROJECT: HOW IT IS BUILT	3 / 4	NTS

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SECTION D · ACCESS

Scan · Explore

Two codes, two ways in. Open your phone camera and point it at either code. Nothing to install.

Explore it in 3D

Scan for the interactive web explorer and AR view.

AUGMENTED REALITY

CODE 1 · AR

The rig, in your room

Place the refrigeration rig on a real table at life size, then switch to the exploded view and walk around it.

01 Scan, then tap **View in your space**.

02 Move the phone slowly over a table or floor.

03 Walk around it; switch to **Exploded**.

Android: Chrome with Google Play Services for AR (ARCore). iPhone/iPad: Safari with AR Quick Look. Other devices get the 3D view. Camera permission is asked only when AR starts.

OR TYPE

shrawan3035.github.io/VCR-AR-Explorer/#/ar

WEB EXPLORER

CODE 2 · WEB

The whole cycle, interactive

Rotate the rig in 3D, tap any of the 12 numbered parts, play the refrigeration cycle and read the component library.

01 Scan to open the explorer.

02 Tap the compressor, condenser, valve or evaporator.

03 Press **Play cycle** or drag **Explode**.

Works in modern browsers on phones, tablets and laptops. Nothing to install.

OR TYPE

shrawan3035.github.io/VCR-AR-Explorer/

Project

TITLE	VCR AR Explorer — Vapour Compression Refrigeration in AR
AUTHORS	[Student names]
GUIDE	[Guide / Faculty name]
INSTITUTION	[Institution / Department]

References & credits

- C. P. Arora, *Refrigeration and Air Conditioning*, McGraw-Hill.
- W. F. Stoecker & J. W. Jones, *Refrigeration and Air Conditioning*, McGraw-Hill.
- Y. A. Çengel & M. A. Boles, *Thermodynamics: An Engineering Approach*, McGraw-Hill.
- Software: three.js, Google <model-viewer>, React, Vite, Unity 6 (URP, glTFast, AR Foundation, ARCore), Khronos glTF-Validator. Fonts: Archivo, IBM Plex Mono (OFL).
- 3D model: original, generated procedurally by this project — educational visualisation, not CAD.

Both codes use high error correction and were test-decoded from the rendered PDF before printing.

TITLE	SHEET	SCALE
SCAN · EXPLORE	4 / 4	NTS